

Development of **AI Data Platform** for Autonomous Ship



Geun-Tae Yim

Director

Autonomous Ship Verification & Evaluation Research Centre
Korea Research Institute of Ships & Ocean Engineering

KRISO

Nor-Shipping 2025

The 6th International Ship Autonomy and Sustainability Summit

Topic

Is AI seen as a problem or a solution? Where do we get reliable training data from?

In Favor

• Sweden (Jacob Terling)

"Sharing high-quality data is key to the advancement of AI"

- If data is treated exclusively as a source of "competitive advantage" and handled in a **closed manner**, it will ultimately create a fragmented system that is only accepted within each country's own waters.

• Korea (Jin Kim)

"As maritime transport is inherently international, the absence of harmonised standards will lead to confusion"

- If AI technology is over-protected or overly tied to national competition, **practical problems** will arise, including insurance and liability confusion.
- There is an urgent need for international cooperation, **led by IMO**, to establish clear standards for **AI system evaluation methods**.

Neutral

• United States (Sean Pribyl)

"In the US, data sharing is constrained by commercial philosophy, trust, and liability issues"

- The focus is on **fair competition** and control of technology, and government platforms are also subject to strict regulation regarding operational information.
- Under GDPR and other data protection rules, companies also tend to take a passive approach.
- Since there are many legal liability issues to resolve, it would be more realistic to begin with voluntary action and **gradually move towards international agreements**.

Against

• Norway (Øyvind Smogeli)

"Sharing data with competitors is a utopian idea and a betrayal of shareholders"

- Instead, a practical alternative was suggested: using functionally isolated training data as a **modular approach**.
- Developers should focus on development itself, while system performance could **be pursued through shared APIs and standardised interfaces**.

• Japan (Hideyuki Ando)

"Data sharing is not possible; cooperation is essential to improve cost efficiency"

- Private companies have invested in collecting data, so the data cannot simply be shared free of charge.
- International cooperation is needed to **improve the cost efficiency of AI training**.
- Cooperation between companies and governments is needed in areas such as clearly defining the functional blocks of navigation systems and **establishing standardised protocols between those blocks**.



"In particular, it seems that everyone agrees that the shortage of data is the single biggest core challenge."

02. Project Overview

3

Title

Development of AI Data Platform for Autonomous Ship

To build a public training dataset to support the development of autonomous ship technologies



Total Budget



Period

2026 ~ 2029 (4 years)

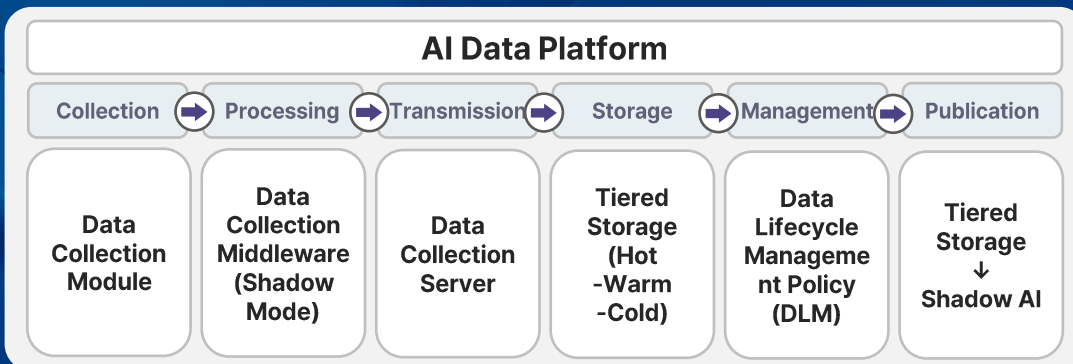


Lead Ministry

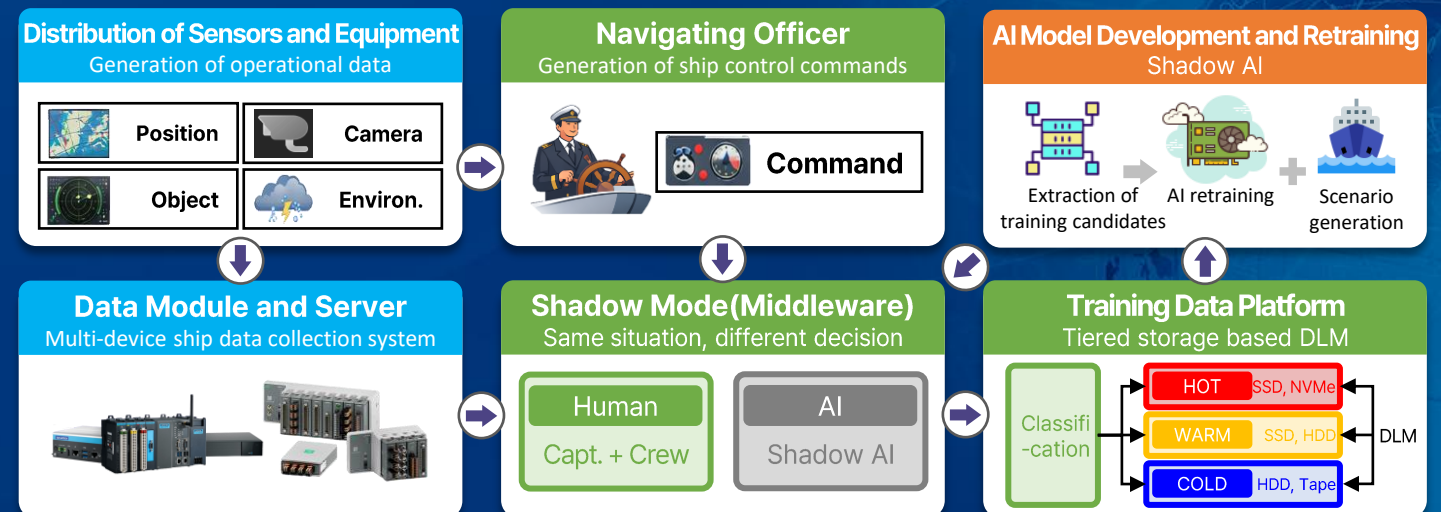
Ministry of Trade, Industry and Resources

Technical Definition and Scope of the AI Data Platform

- **(Definition)** A system that processes ship operation data throughout the full lifecycle — collection, processing, transmission, storage, management and publication — in order to generate and distribute training datasets for AI models
- **(Scope)** Onboard data collection → Shadow Mode-based processing → data transmission → tiered storage → data lifecycle management → use for AI training, verification and publication



- **(Concept)** A structure for AI model retraining, updating and continuous improvement



02. Project Overview

Development of AI Data Platform for Autonomous Ship

(Sub 2) AI Training Data Platform

- Tiered storage (Hot-Warm-Cold)
- International standardization (IMO, ISO etc.)

(Sub 3) AI Model & Service

- Development of Shadow AI(DataOps, MLOps)
- Public training datasets

(Sub 1) Supply & Acquisition

- 30+ vessels/fleets
- Installation of data collection platform

Candidate training datasets

Tiered Storage
(Hot-Warm-Cold)

AI Training
Data Platform

HOT (High Importance)

WARM (Medium Importance)

COLD (Low Importance)

Shipping companies,
Equip. manufacturers

Data Lifecycle
Management

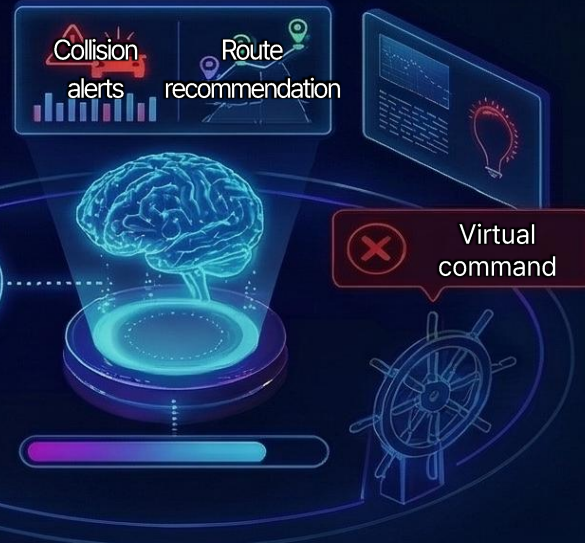
Academia,
Shipbuilders etc.

Update

Shadow AI Release

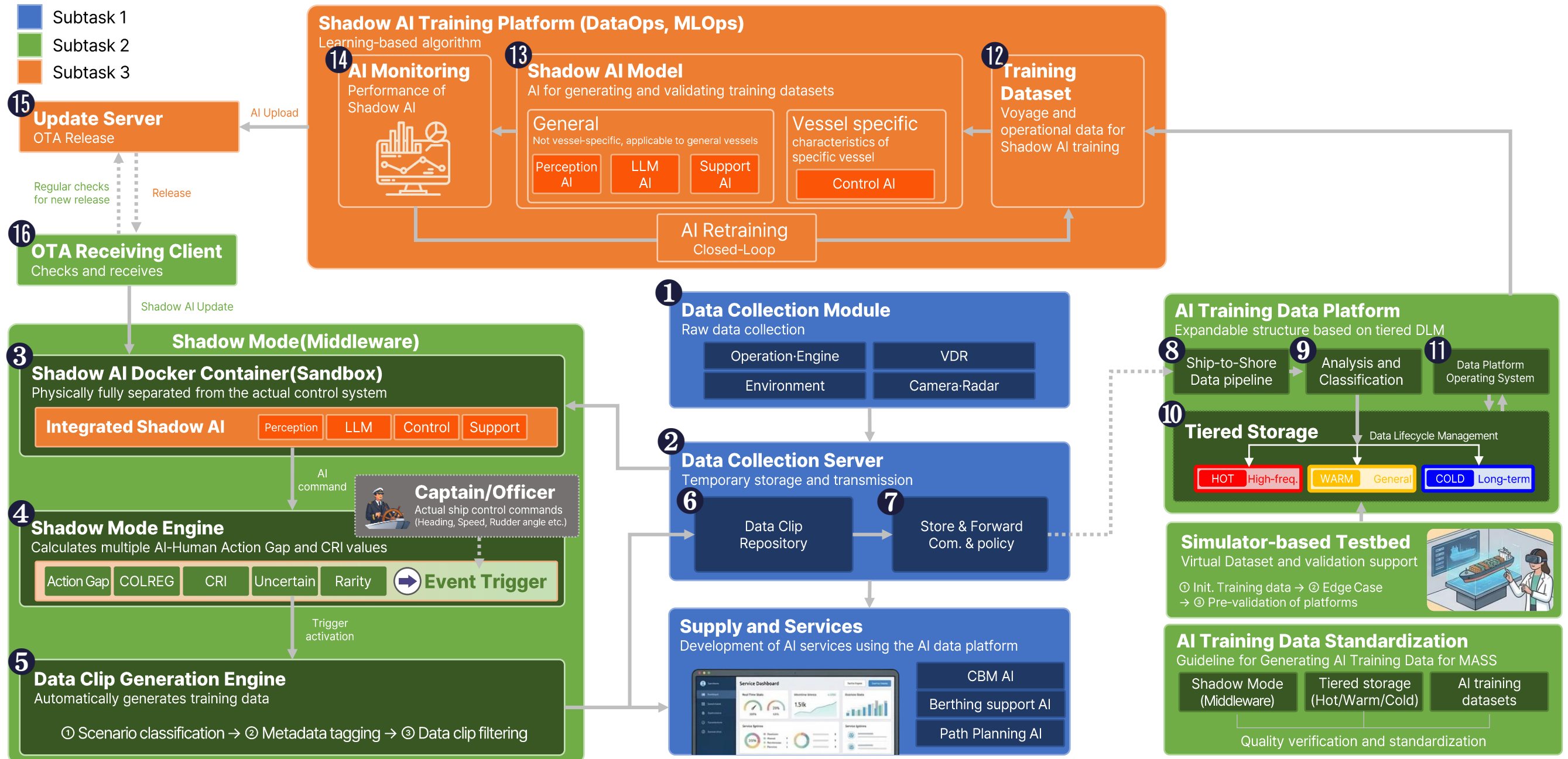


In operation:
Shadow AI



03. Concept Diagram

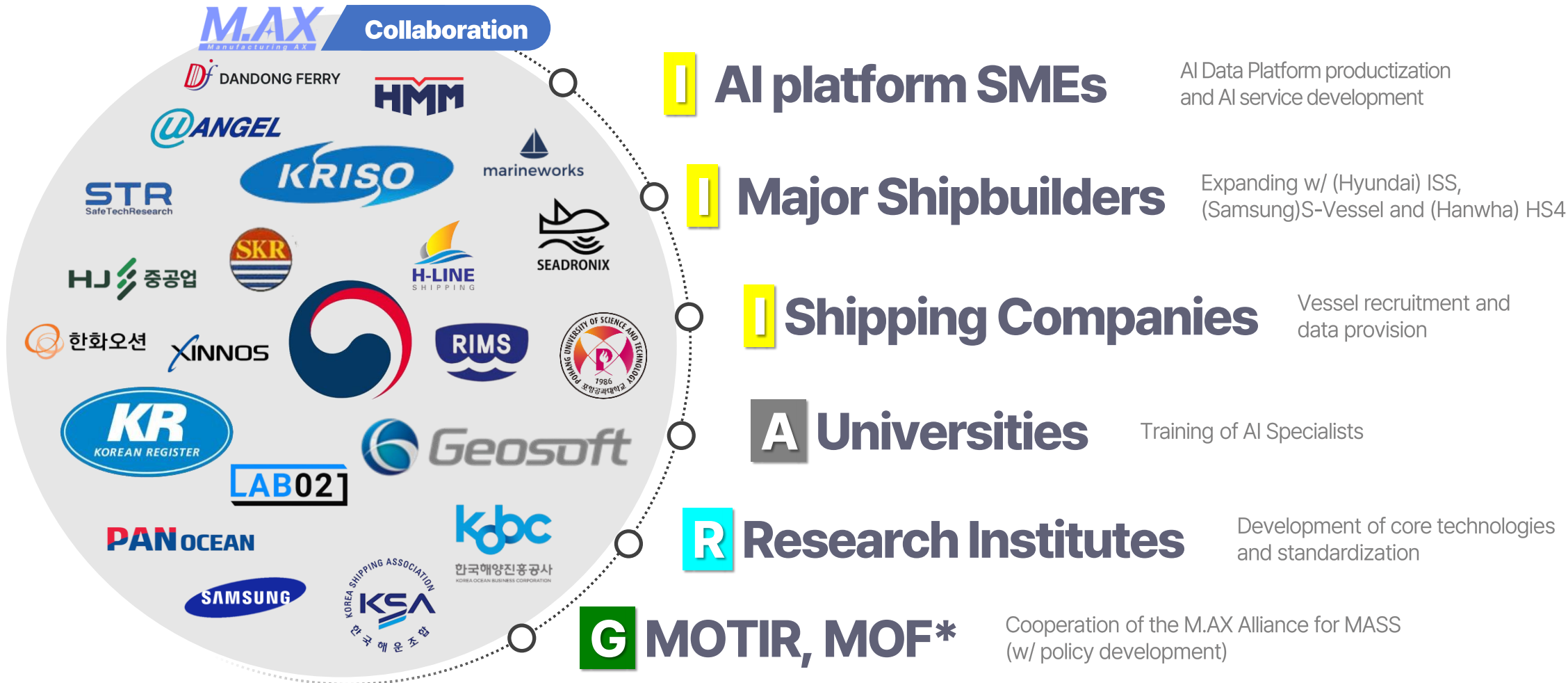
5



04. Strategy

6

① Project Consortium: industry, academia, research, and government for maritime AI



I: Industry, A: academia, R: research, G: government
MOTIR: Ministry of Trade, Industry and Resources
MOF: Ministry of Oceans and Fisheries

Application of Outputs from the 「Korea Autonomous Surface Ship(KASS)」

*The image above may differ from the actual research findings.

③ Collaboration with Private Organizations and Data Publication

Ensuring Technology Scalability through Collaboration with Maritime Industry

Shipbuilders



Shipping Company



Public Org.

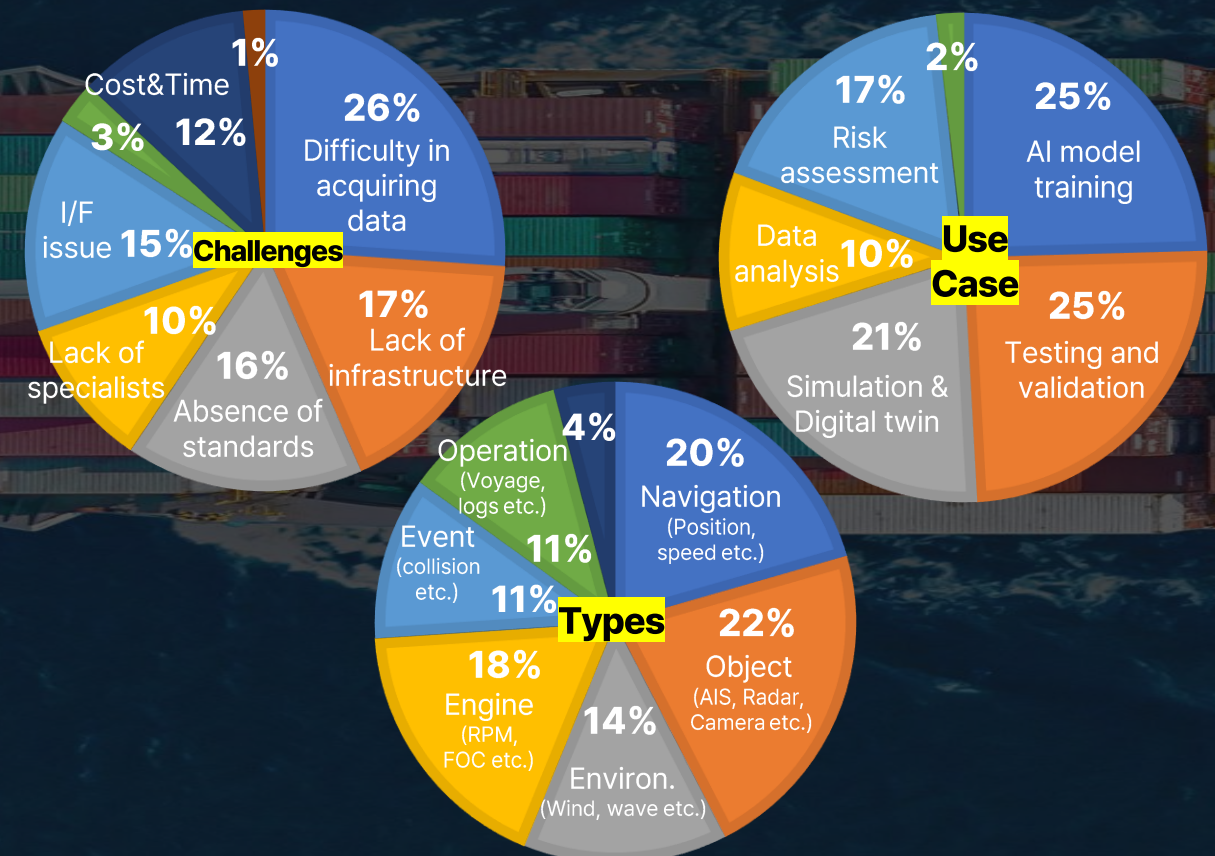


Private Org.



Public Release of Proprietary Maritime Datasets

Industry demand was confirmed through a stakeholder survey



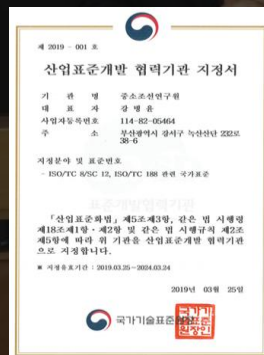
④ Development of International Standards



Capability

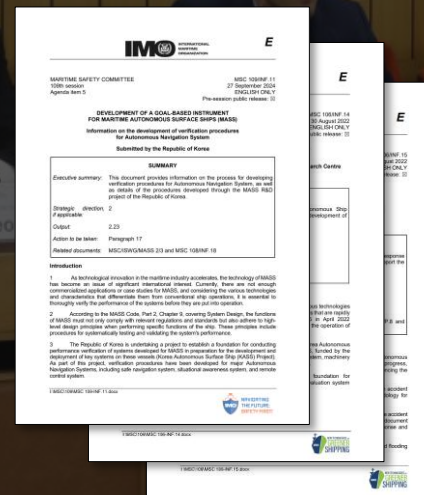
To standardize project outcomes to international standards

KRISO Specialized institution for ship and shore R&D
KR Member of IACS
RIMS Certified as COSD



Standardization Activities (IMO-ISO)

- ▲ International standards development
- ▲ AI standardization TF participation
- ▲ Forum and working group activities etc.



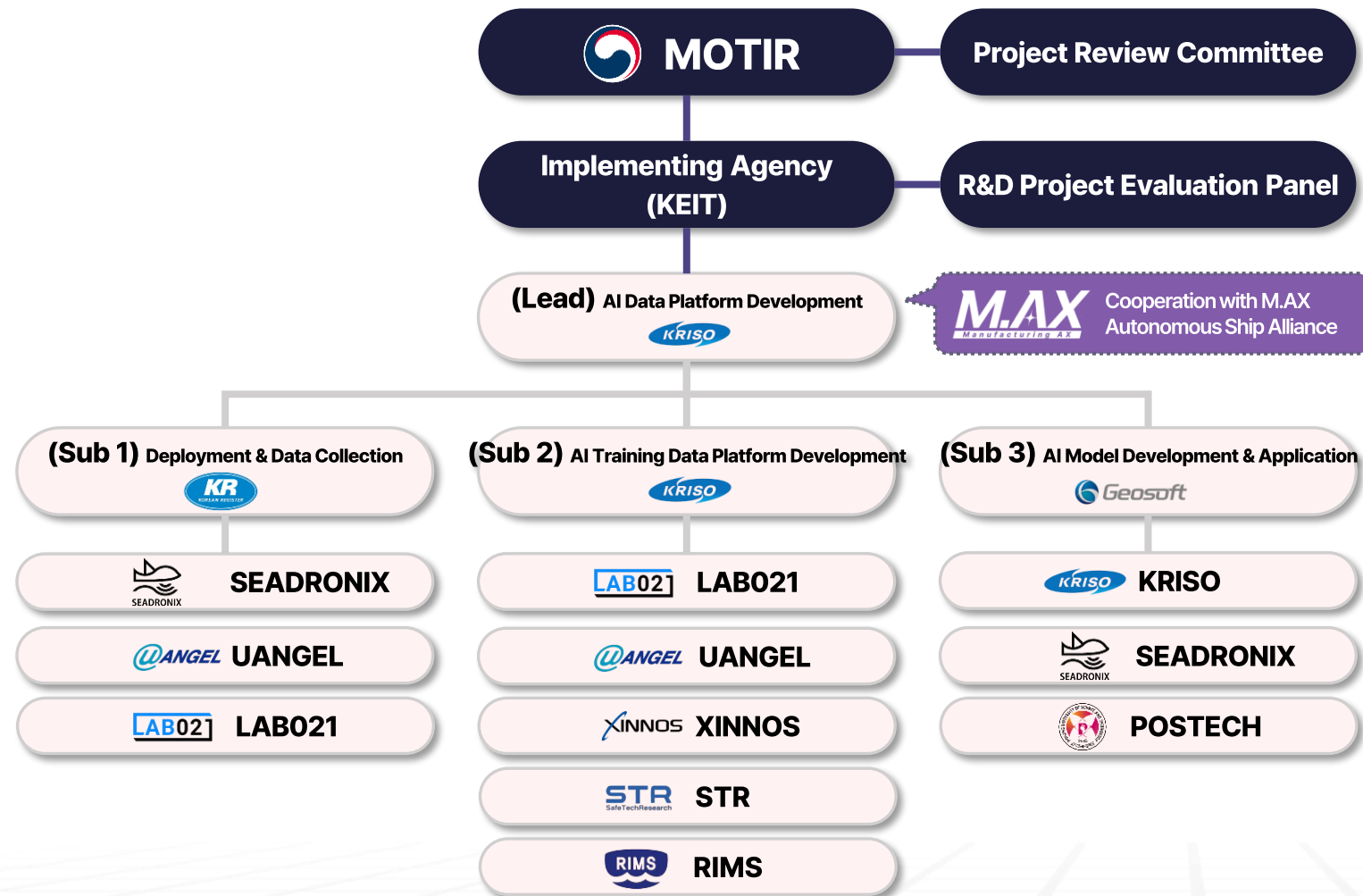
International Cooperation

- ▲ Promoting Korean technologies through International symposia and exhibitions (e.g. SMM, Nor-Shipping)
- ▲ Global cooperation with leading-countries (e.g. Norway, UK, Japan)



05. Project Structure

10



Development R&R		
Cat.	Org.	Role
Lead	KRISO	Overall project coordination
Sub1	KR	Vessel securing and onboard infrastructure deployment
	SEADRONIX	SA sensor system design and AI service linkage
	UANGEL	Onboard data pipeline and operation system development
	LAB021	Data acquisition module and server S/W development
Sub2	KRISO	AI platform algorithm development and operation
	LAB021	Shadow Mode middleware development and deployment
	UANGEL	Tiered storage and dataset management development
	XINNOS	Ship-shore pipeline and AI platform operation system development
	STR	Simulation-based synthetic data generation and verification
	RIMS	AI dataset standardization research
Sub3	Geosoft	Shadow AI autonomous navigation model development
	KRISO	Ship-motion-based AI control and V&V scenario development
	SEADRONIX	Multisensor SA AI model development
	POSTECH	Ontology-LLM maritime reasoning AI development

01. Target Vessel Selection Criteria

Coastal Vessels



Fishing Vessel



Car Ferry



Government Vessel



Port Service Vessel

Characteristics	Irregular route, small vessel, poor maritime communication	Fixed routes, Frequent port calls	Mission-based operation, Latest standardized equipment	Dense coastal operation, low speed
Strength	Extreme environment, Irregular maritime data	Ultra-high-frequency port calls, reliable fixed sections	High-resolution sensors, Stable maintenance	Ultra-dense port traffic
Limitation	Limited space, Frequent communication shadows	Limited operating area	Data security regulations	Limited high-speed operation
		✓	✓	✓

Ocean-going Vessels



Container Ship



Bulk Carrier

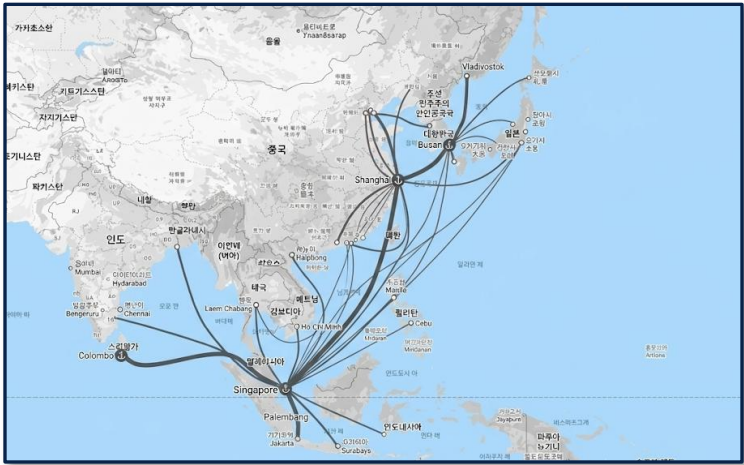


Tanker

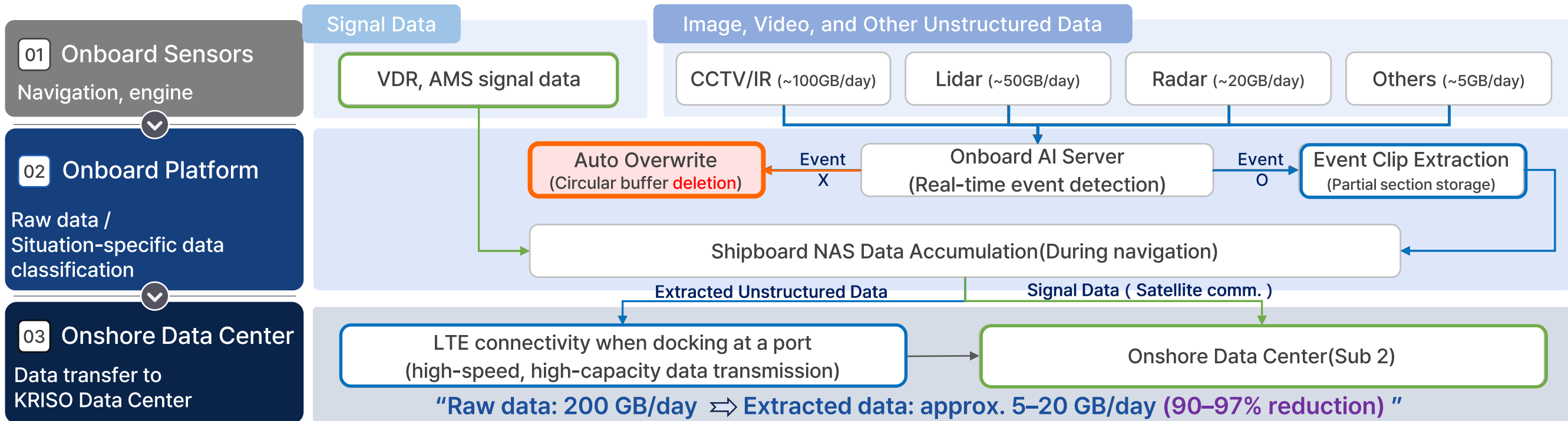
Asia Route Data

Short-distance, Coastal, and regional routes

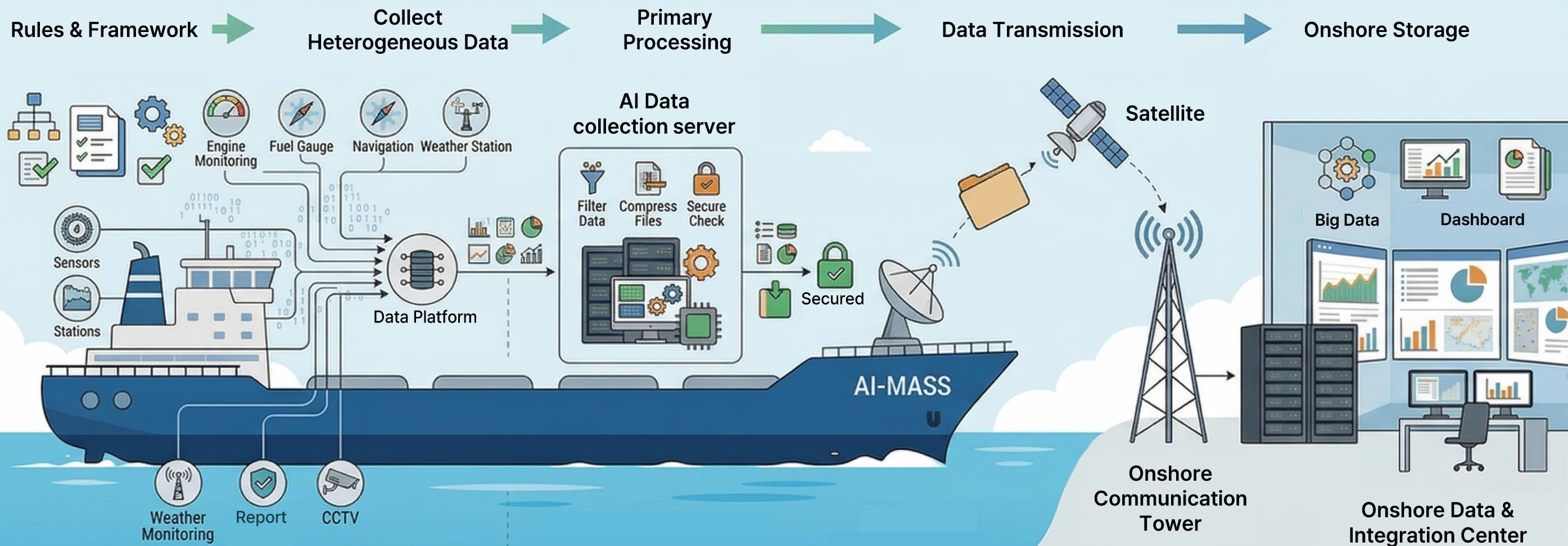
- Feeder container or similar-class vessel
- For large container ships, ocean-going data may be relatively monotonous due to autopilot-centered operation



02. Data Acquisition Strategy



03. Development Contents



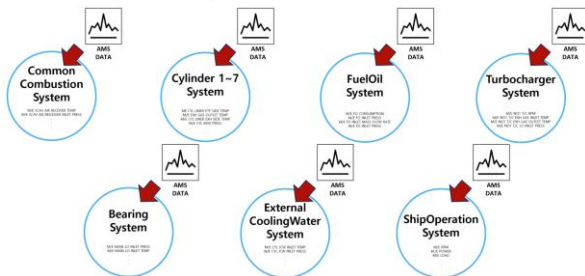
04. Onboard AI Services

01

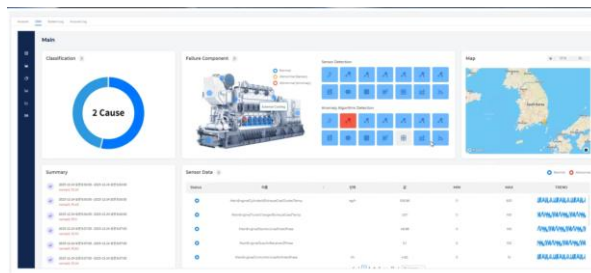


Machinery Predictive Maintenance AI Service

- Collect and preprocess machinery data from target vessels



- Tune predictive maintenance AI algorithms
- Apply and enhance CBM AI service on real vessels

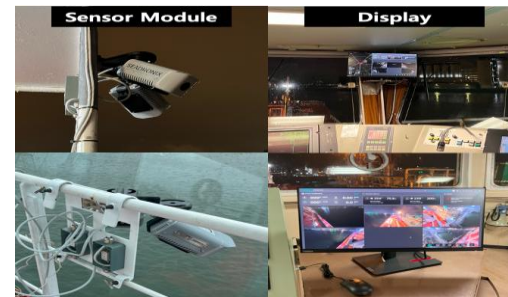


02



Route Support AI Service

- Collect and preprocess SA data from target vessels

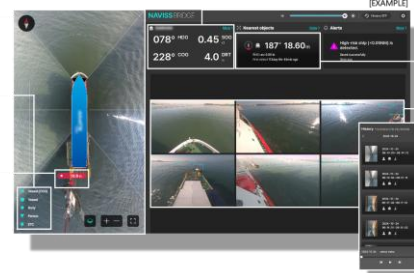


- Tune route support AI algorithms
- Apply and enhance route support AI service on vessels

Real-Time 360° Around View
Provides stabilized and seamless real-time video by combining multiple sensors. Offers zoom-in/out functionality.

Target Object Classification
Automatically detects and classifies objects, even in obstructed visibility conditions.

Navigation Status Information
Automatic recognition of all vessels and obstacles around, including vessels with AIS deactivated or undetected by radar. Analysis of distance information for identified obstacles.



AIS Integrated Information
Displays real-time navigation status.

Proactive Risk Alerts
Alerts for potential collision risks with nearby objects through analysis.

Customizable Mode
Offers user-specific UI customization, such as wide single-view and composite views.

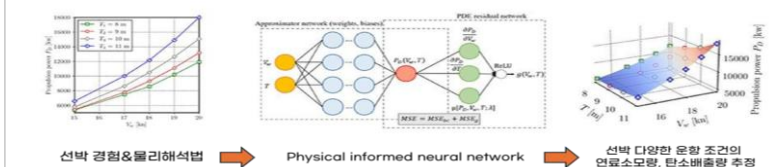
Operation History
Provides real-time saved operation recording video.

03

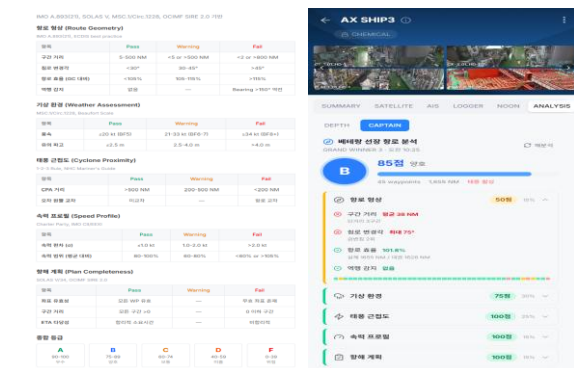


Voyage Plan Auto-Diagnosis AI Service

- Develop PINN-based fuel consumption prediction model



- Develop voyage plan diagnosis rating algorithm
- Apply and enhance voyage plan auto-diagnosis AI service on real vessels



01. Subtask Overview

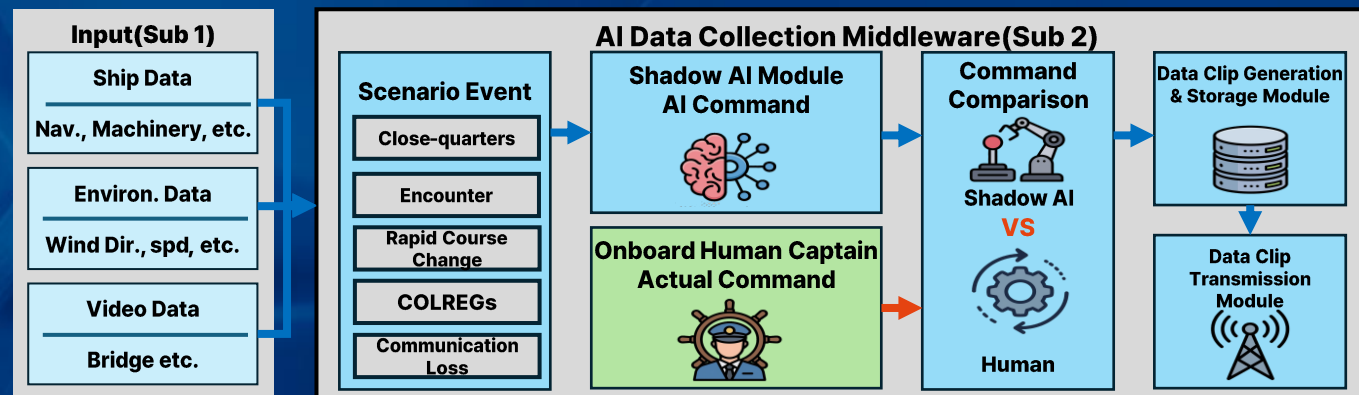
Sub 2

Development of an AI Training Data Platform for MASS

Development of Core Technologies for Generating High-Quality AI Training Datasets for MASS

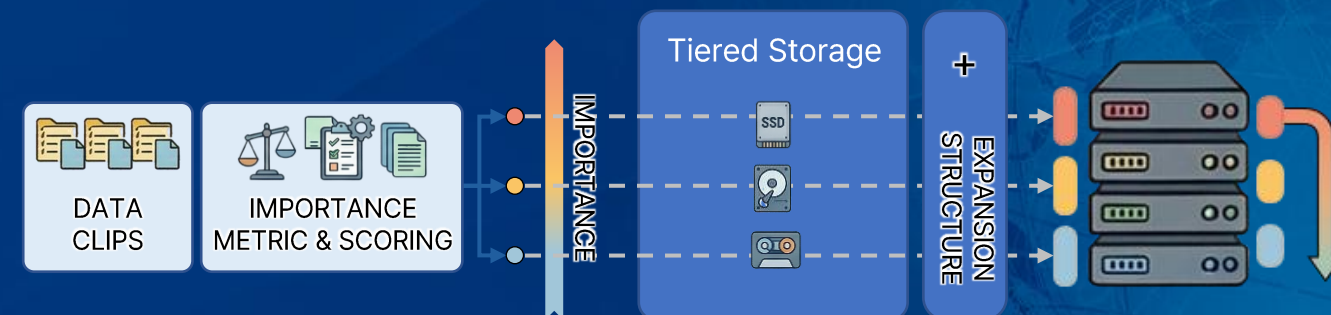
| AI Data Collection Middleware

- ➔ **(Definition)** An onboard AI data pipeline that collects and preprocesses operational data in real time, clips and packages data for training purposes, and uploads it to the AI data platform
- ➔ **(Scope)** ① AI model background prediction → ② Shadow Mode → ③ Data clip upload + ④ Additional function support



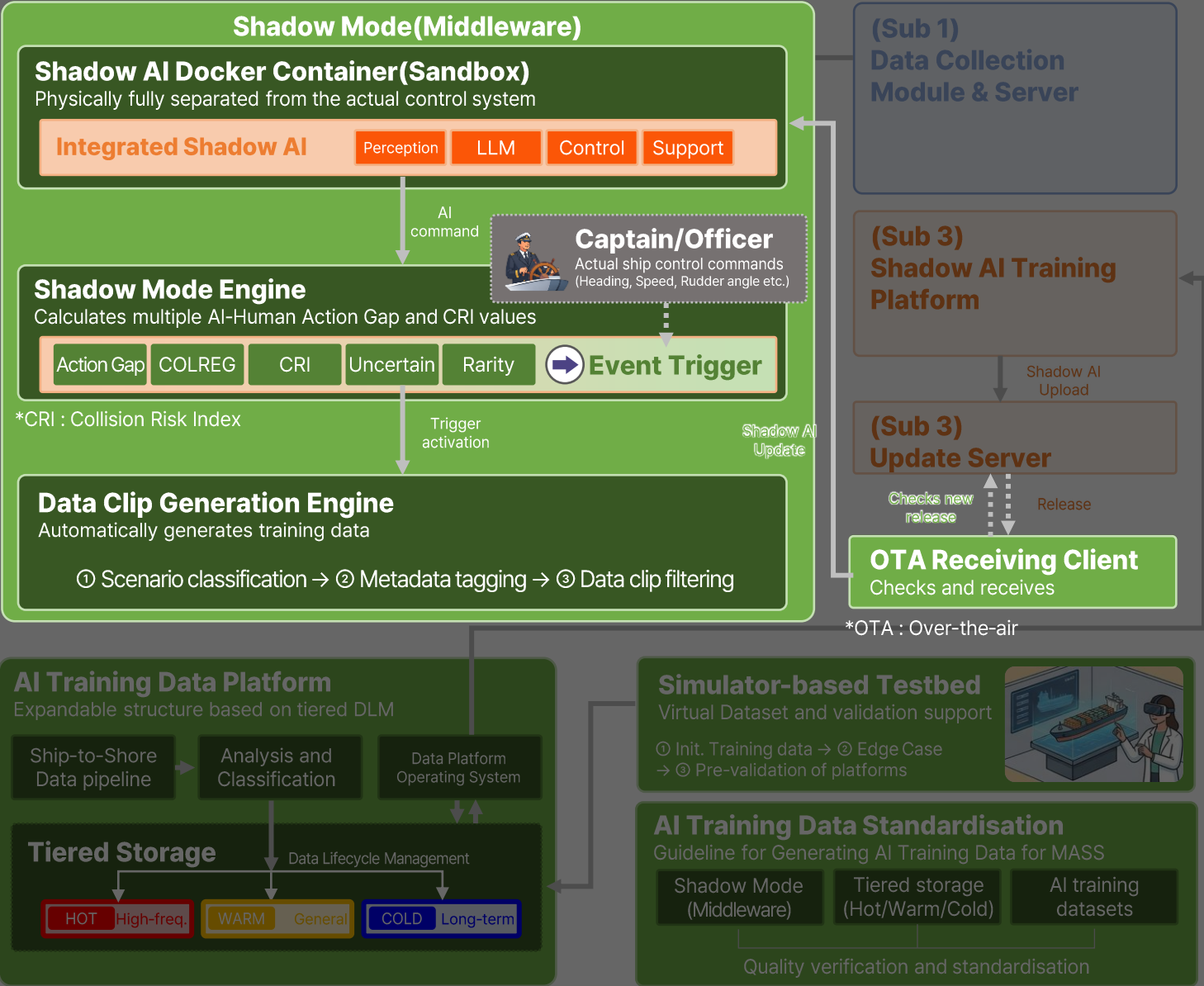
| Tiered Storage and AI Training Data Platform

- ➔ **(Definition)** A data platform that analyzes the training value of large-scale data clips transmitted from the AI data collection middleware, stores them in a tiered structure, and supports dataset management
- ➔ **(Scope)** ① Data clip value analysis → ② Tiered storage → ③ DLM → ④ Deployment



02. Details ①: AI Data Collection Middleware

Sub 1 Sub 2 Sub 3



Key Issues

- ① 24/7 unmanned stability in the maritime env.
- ② Undefined trigger threshold
- ③ Need to secure quality of training data

Development

Shadow Mode-based Sandbox Environment

A physically isolated test environment from the actual ship control system, where AI commands are generated only as virtual commands and are not reflected in actual steering/control

AI-Human Action Gap

Analysis of differences between the Shadow AI model and the actual navigator/human commands and decisions
→ Implementation of “Learn from Human”

Input (Sub 1)

- Ship Data: Navigation, engine
- Environmental Data: Wind dir./spd., temperature
- Video Data: Bridge, engine room etc

AI Data Acquisition Middleware (Sub 2)

- Scenario Event Model: Close encounter, Encounter, Course, COLEGS, Lost comm.
- Shadow AI Control Module: AI Control Command
- Onboard Captain Actual Control Command
- Command Comparison Module: Shadow AI vs Human
- Clip Data Generation & Storage Module
- Clip Data Transmission Module

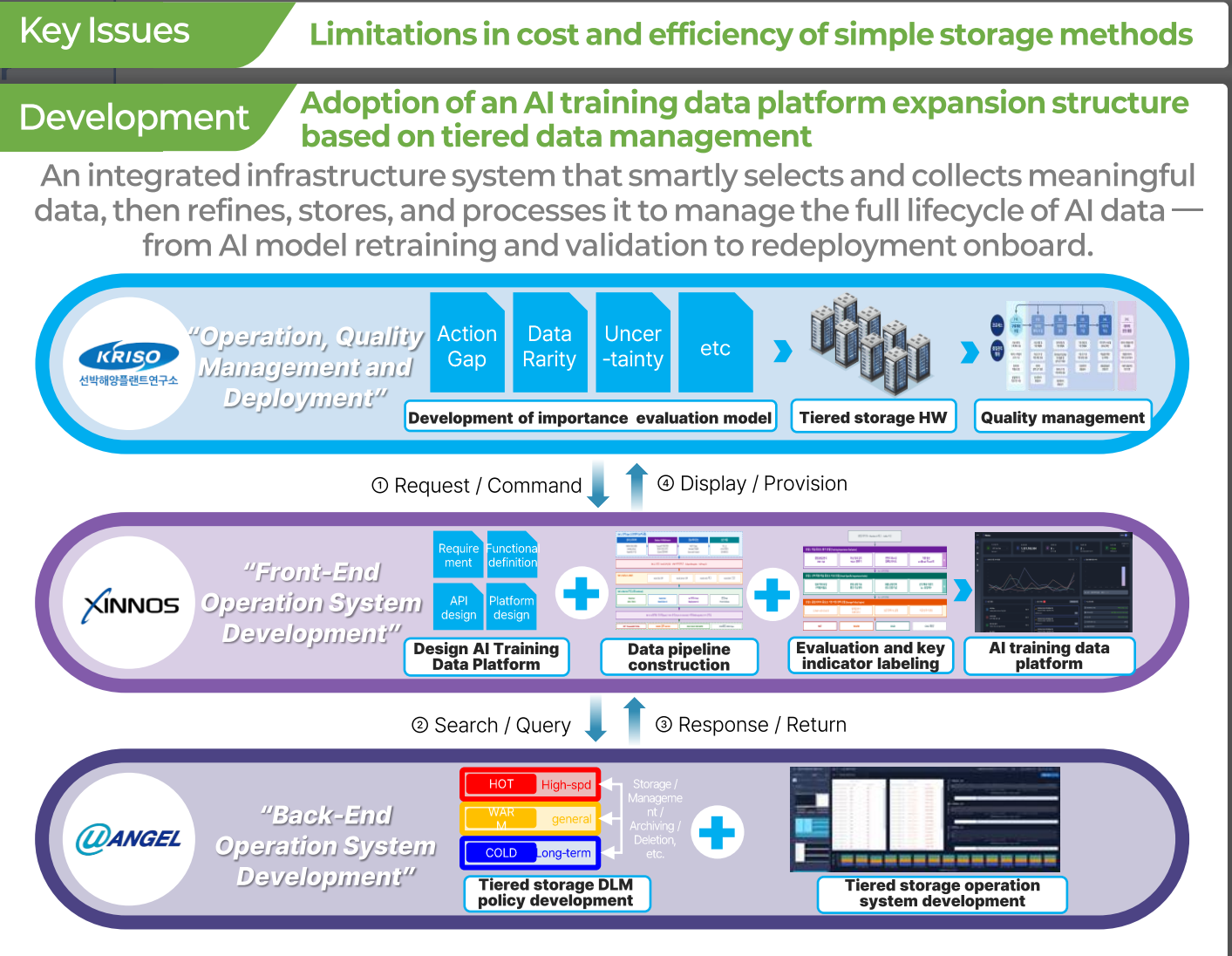
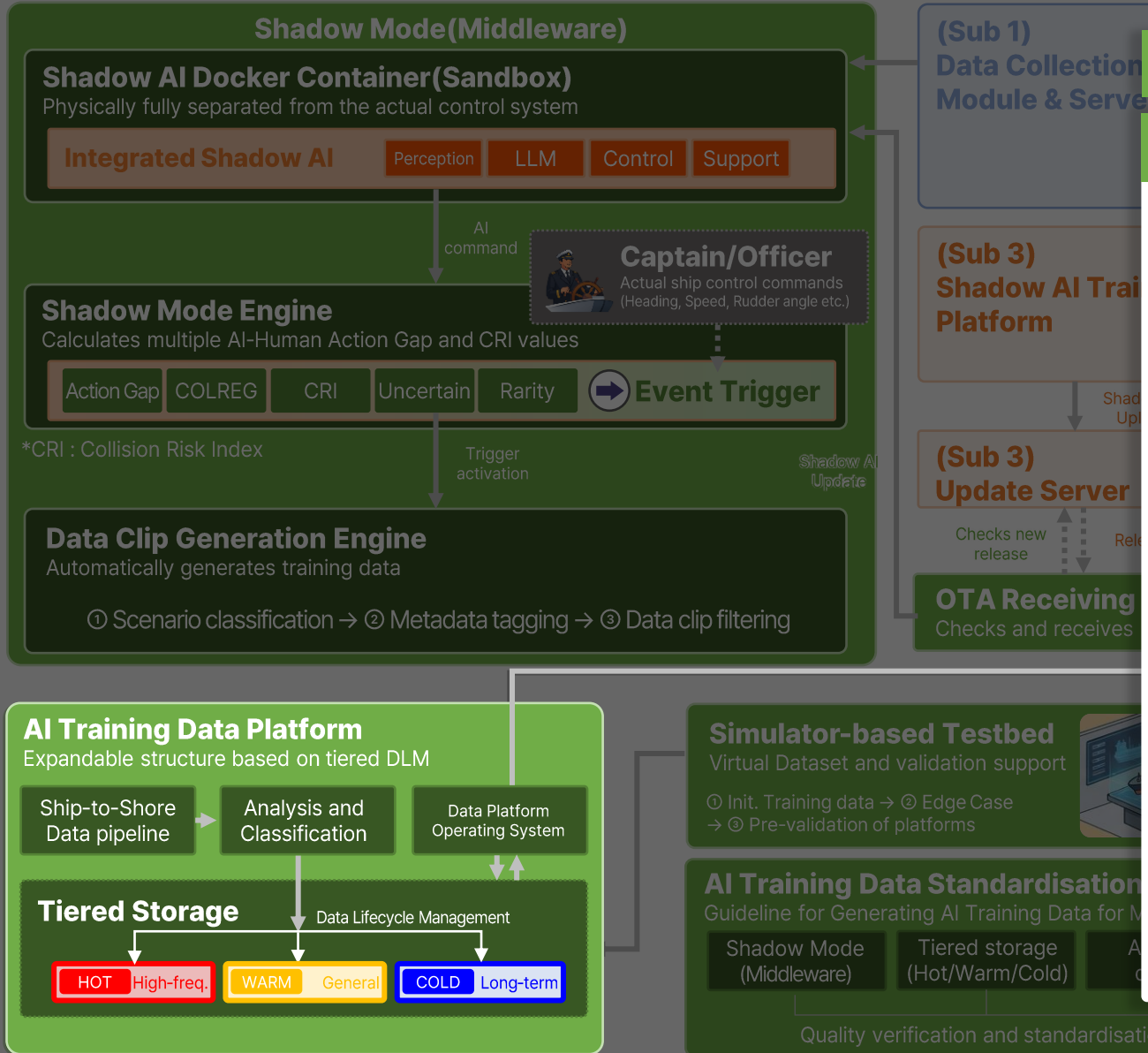
HUMAN vs AI

Action Gap: AI vs. Human, AI vs. Decision

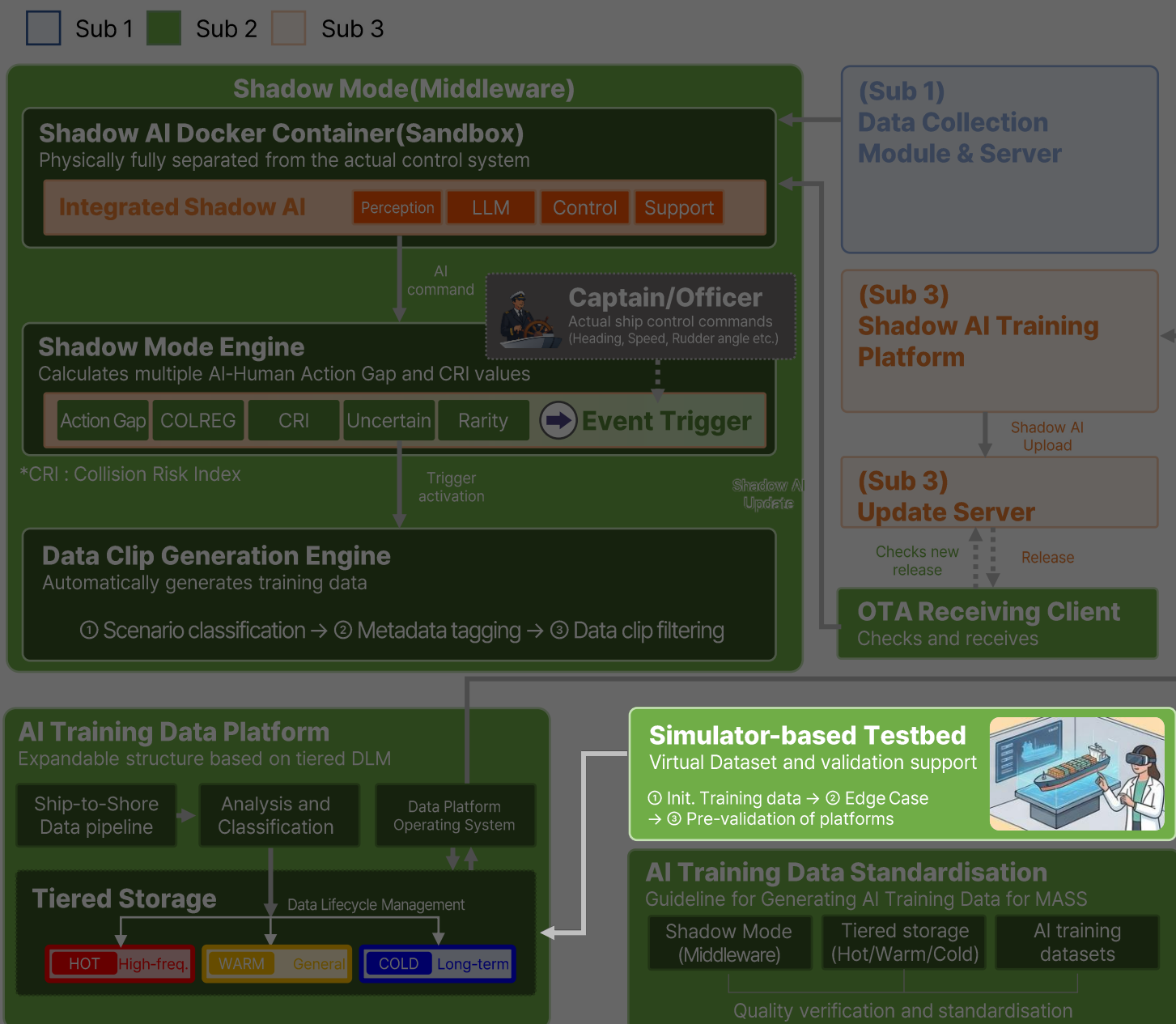
CRI: Uncertain, Rarity

02. Details ② : Tiered Storage & AI Training Data Platform

Sub 1 Sub 2 Sub 3



02. Details ③ : Simulation Data-based Technical Support



Key Issues

Lack of initial data and integrated verification environment

Development

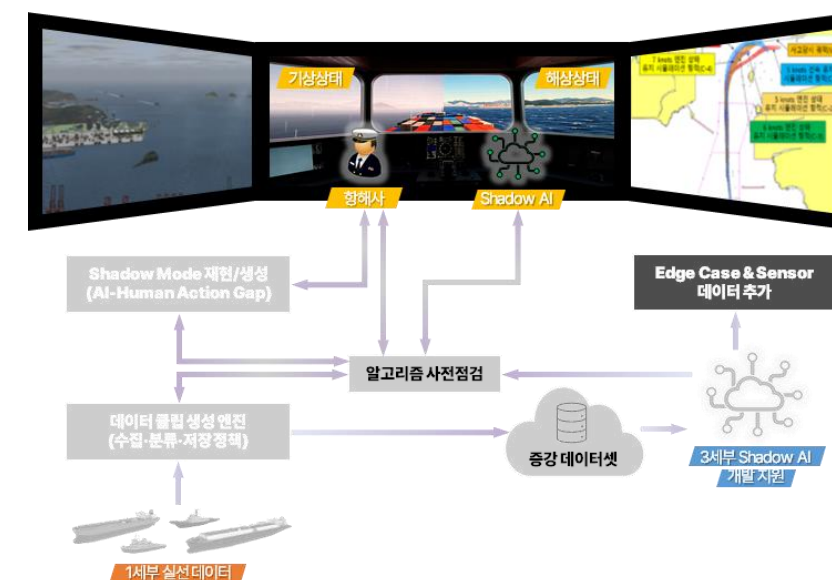
Support AI training and validation using simulation data

Initial Data & Pre-validation

Build initial training data and pre-validate Shadow AI/data middleware before onboard deployment

Edge-case Data Generation

Generate rare and hazardous edge-case scenarios that are difficult to obtain in real operations



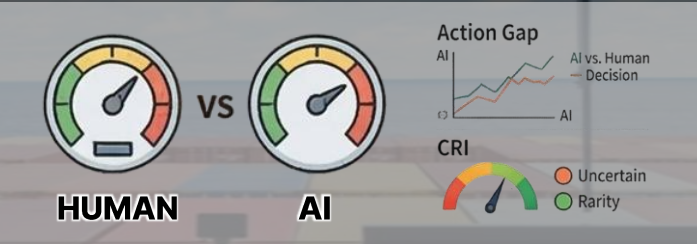
03. Data Strategy (Collection → Management → Publication)

Existing operational dataset held by shipping companies

Ship data integration & exchange middleware development



Shadow Mode dataset



Virtual dataset



Coll.

Mgmt. Pub.

2026 Initial testing & validation

Data pipeline interfaces between subtasks



2028 Data lifecycle management

Development of DLM policies, including data-value-based storage, automated tiering, and deletion

Action	Trigger condition	Path	Update Metadata (Arag workflow)
Ingestion	New dataset (Write)	External → Hot	ter=HOT, last_access=NOW
Warm-down	last_access > 30 days	Hot → Warm	ter=WARM, edit endpoint
Archiving	last_access > 90 days	Warm → Cold	ter=COLD, is_archived=True
Purge	Expiration period (e.g., 3 years)	Cold → Deletion	DB Record Deleted
Promotion	last_access < 2 days	Warm → Hot	ter=HOT, edit endpoint

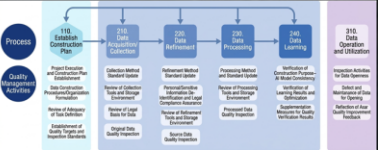
2027 Data management policy

Before tiered storage → On-shore repositories
After tiered storage → Access control policy

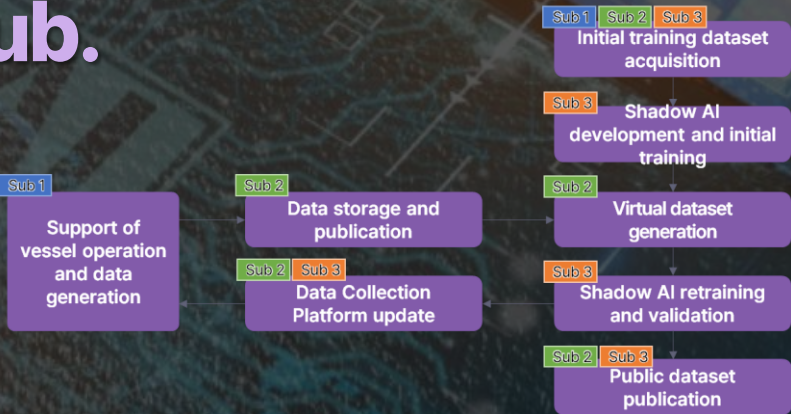
Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy
Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy
Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy
Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy
Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy
Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy	Access Control Policy

2029 Data quality management

AI training data quality process
ISO-based compliance & certification



- ① Definition of target industries
Shipbuilders, shipping companies, equipment manufacturers, start-ups etc.
- ② Survey
Survey and interviews on required dataset items, formats, and use cases
- ③ Generation of public datasets
Identification of priority datasets and generation/validation
- ④ Public dataset publication policies
Scope of disclosure, access rights, terms of use, security, and publication methods
- ⑤ Publication plan
Benchmarking existing platforms (e.g. AI Hub, Waymo)
- ⑥ Operation of public data platform
Support for target industries to use public datasets



03. Data Strategy (Infrastructure)

Real-Sea MASS Data Hub

ASVERC



Ship & Ocean Data Center

**Ship & Ocean
Digital Twin Center**



Public Training Data Access

Public Cloud Service



01. Subtask Overview

Sub 3

Development and Application of Physical AI Models for MASS

Develop learning-based autonomous navigation AI models(Shadow AI) based on autonomous navigation algorithms, and establish a scenario-based safety verification and retraining framework aligned with international standards

Content

Development of Learning-Based Autonomous Navigation Shadow AI Model

Development of Autonomous Navigation Shadow AI

- ➔ Develop initial Shadow AI model and integrate with vessel systems
 - SA + Non-Intervention Control + LLM/Ontology + Decision Support
- ➔ Secure virtual data based on new and complex scenario simulations

Establishment of Shadow AI Operation Platform

- ➔ Design AI model training infrastructure and architecture
- ➔ Build repeated learning framework based on AI data platform
- ➔ Develop and establish AI operation platform

Establishment and Deployment of Public AI Datasets by Industry Sector

Securing, Deploying, and Utilizing Public AI Datasets

- ➔ Develop verification and certification scenarios for autonomous ships
- ➔ Identify and refine demand for MASS datasets
- ➔ Support public dataset release, deployment policy, and utilization

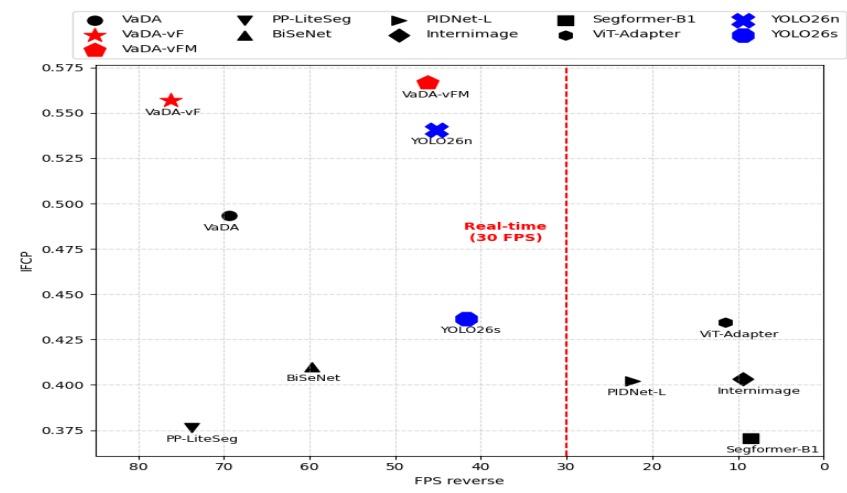
02. Shadow AI Development – Situational Awareness

Situational Awareness AI

Applicable to various vessel types as a SOTA foundation model for maritime situational awareness based on sensor fusion



Performance of In-house Foundation Model VaDA



Example Results of Maritime SA Model



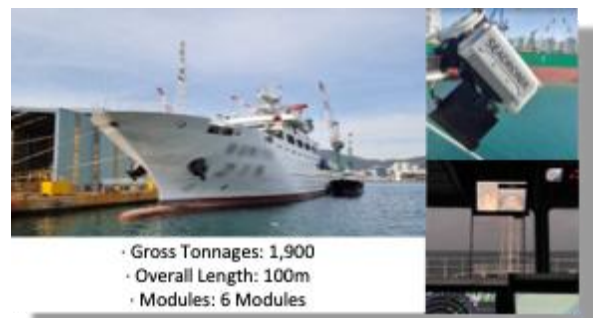
Example Applications to Various Vessel Types



- Gross Tonnages: -
- Overall Length: 60m
- Modules: 4 Modules



- Gross Tonnages: -
- Overall Length: 30m
- Modules: 4 Modules



- Gross Tonnages: 1,900
- Overall Length: 100m
- Modules: 6 Modules



- Gross Tonnages: 3,700
- Overall Length: 100m
- Modules: 5

Autonomous Navigation Vessel

CTV

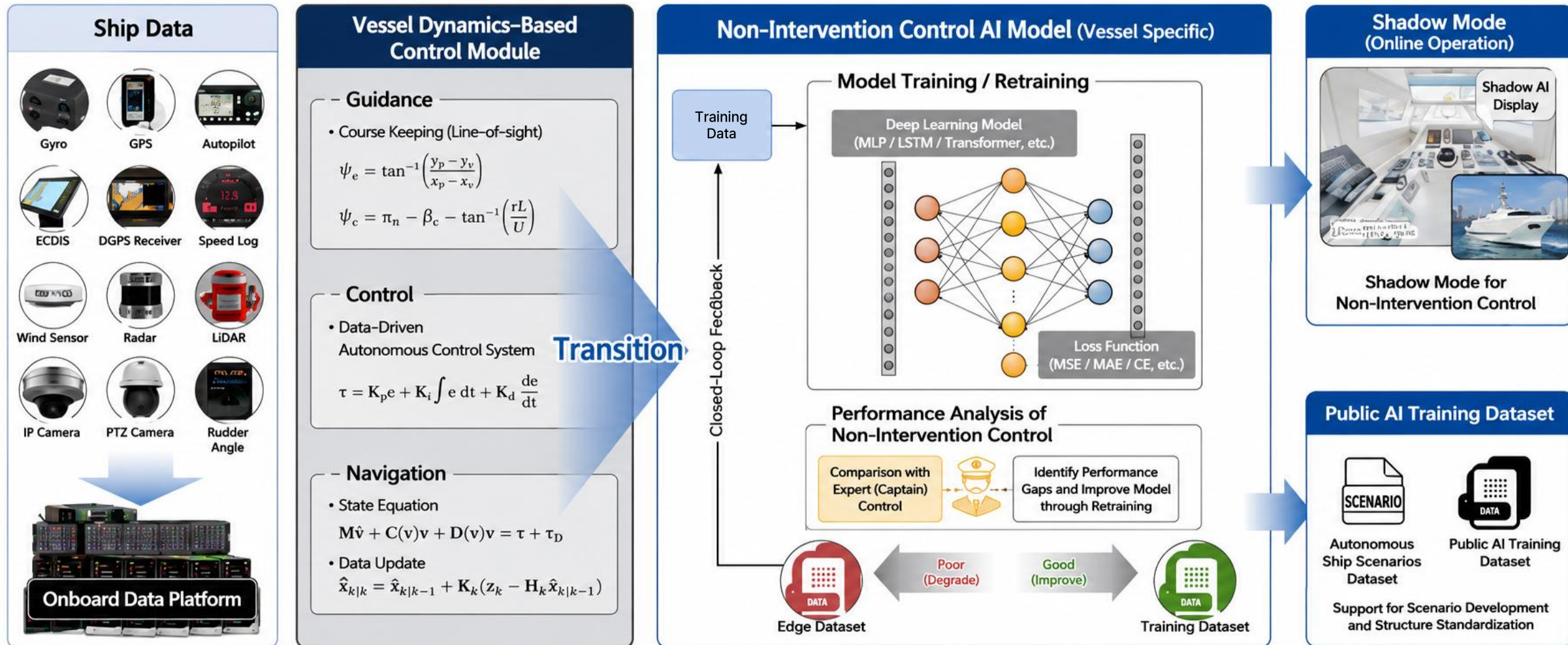
Fishery Monitoring Vessel

Oil Tanker

03. Shadow AI Development – Non-intervention control

Non-intervention
Control

Control command output based on ship motion and autonomous navigation data



Gyro: Gyroscope GPS: Global Positioning System ECDIS: Electronic Chart Display and Information System
DGPS: Differential GPS LiDAR: Light Detection and Ranging IP Camera: Internet Protocol Camera
PTZ Camera: Pan-Tilt-Zoom Camera

ψ_e : Line-of-sight Course ψ_c : Desired Heading r : Yaw Rate L : Ship Length U : Ship Speed
 τ : Control Input e : Error K_p, K_i, K_d : PID Gains
 $\mathbf{M}$: Inertia Matrix $\mathbf{C}(\mathbf{v})$: Coriolis Matrix $\mathbf{D}(\mathbf{v})$: Damping Matrix $\mathbf{v}$: Velocity Vector $\boldsymbol{\tau}_D$: Disturbance

MLP: Multi-Layer Perceptron LSTM: Long Short-Term Memory
MSE: Mean Squared Error MAE: Mean Absolute Error
CE: Cross Entropy

KASS* 2nd Project

*Korea Autonomous Surface Ship

01. Overview

Project

AI Fully Autonomous Vessel Technology Development Project

Achieve IMO Level 4 through fully autonomous navigation technology combined with AI



Total Budget

KRW 464 billion



Period

2026 ~ 2032 (7 years)



Lead Ministry

Ministry of Trade, Industry and Resources

Ministry of Oceans and Fisheries

Definition and Scope of AI Fully Autonomous Ships

- ➔ **(Concept)** Beyond the current level of integrating advanced technologies like AI into navigation systems to support crew decision-making, to a level where crew intervention is minimized for operational capability

Existing Vessel

Ship + Crew

Autonomous Vessel

Ship + Autonomous Navigation System(Navigation/Engine) + Remote Operation Center(Land-based)



Core Elements: Ship Systems, Safety, Crew Onboard and Degree of Autonomous/Remote Operation, Cybersecurity/Communications, etc.

- ➔ **(Classification)** Distinguished by automation level: Partially Autonomous (Lv1~3) and Fully Autonomous (Lv4) (IMO)

Level 1

Crew Decision Support



Level 2

Crew boarding, Remote Control



Level 3

No Crew Onboard Remote control



Level 4

Fully Unmanned Autonomous Vessel



03. Vision Framework

Purpose



Leading the global autonomous shipping market through AI autonomous navigation technology



Goal

Achieve IMO Level 4 through fully autonomous navigation technology combined with AI

Performance Goals

Securing Core AI Autonomous Navigation Technologies

- Registered patent SMART

Advancement of AI Fully Autonomous Unmanned Systems

- Performance targets for unmanned systems (%)
- Establishment of global inspection and certification frameworks

Leadership in International Standards for AI Fully Autonomous Ships

- MASS international standardization response

| Project Structure |

Item 1

Development of a Physical AI-Based Unmanned Navigation System

AI-based real-time recognition, decision, and control for unmanned navigation processes

Item 2

Development of Intelligent Engine Automation System for Unmanned Navigation

Safety management, predictive maintenance, control and operation for automated and intelligent systems

Item 3

Development of AI Fully Autonomous Ship Operation Technology

Remote operation, port integration, unmanned cargo management – enabling fully unmanned operation

Item 4

Development of AI Fully Autonomous Ship Verification and Demonstration Technology

Reliability and performance evaluation technologies, including real-sea demonstrations

Fully Autonomous Ship Operations Technology Development

01 Port Connectivity

- AI-based integrated port operation system
- Cargo arrival/departure reporting, customs/immigration/quarantine
- Onboard mooring equipment and automated berthing

04 Seamless Communication

5G, low-earth-orbit satellite, and next-generation communication-based optimized data exchange

05 Unmanned Cargo Operation

Cargo state monitoring, emergency response, and other autonomous operations

02 Development of End-to-End Remote Operation Technology

03 Remote Operation Support Technology

Remote Operation Centre

Departure → Voyage → Arrival



Thank you